

MADDU JAGADESH

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Summary & Career Objective

Highly motivated and results-driven Integrity Engineer with over 2.5+ years of progressive experience in Mechanical Integrity (MI), Quality Control (QC) for piping systems, and process optimization within the Oil & Gas sector. Proven expertise in Pressure Relief Device (PRD) optimization, piping circuitization, PFD preparation, and ensuring asset reliability. Seeking to leverage a strong background in engineering standards and asset management to contribute to a challenging and dynamic organization.

Key Professional Certifications

➤ API 510: Pressure vessel Inspector

- Issued – Jan 2026
- Credentials – 131969

➤ API 580: Risk-Based Inspection (RBI)

- Issued – Aug 2025
- Credentials – 129043

➤ ASNT NDT LEVEL II: RT/UT/MPT/DPT

- Issued – Aug 2023
- Credentials – (2300/2301/2302/2303)/0068

Professional Experience

Mechanical Integrity Engineer (Full-Time)

Quest Global Engineering Services Private Limited | Nov 2024 – Present

Piping Circuitization and preparing the piping – Corrosion loops

- Proficient in developing and analyzing piping circuitization from technical diagrams like PFDs (Process Flow Diagrams), P&IDs (Piping and Instrumentation Diagrams), and Isometric drawings to generate corrosion loops, to assess corrosion risks with applicable damage mechanisms, to develop an inspection & monitoring strategy with a well-defined scope, areas of focused assessment with defined inspection intervals.
- Performed piping mark-ups at the P&ID level using AutoCAD and preparing comprehensive line lists with material, process details utilizing the Data Extraction tool for further streamlined analysis to develop strategies for integrity risk assessment with remaining life calculations.

- Updated and revised isometric drawings to incorporate new design changes and Management of Change (MOC) updates.
- Automated the conversion of AutoCAD files to PDF format using Auto LISP and BAT file scripting, significantly reducing manual effort and optimizing workflow efficiency.
- Partially automated the Master Asset List (MAL) preparation process in Excel, cutting down overall working hours and enhancing operational productivity.
- Developed and designed block diagrams/ Process Flow Diagrams (PFDs) for piping circuits to enhance process clarity and facilitate efficient circuit systemization.

Risk Based Inspection - Equipment Strategy

- Developing Equipment Strategies for Fixed Equipment's and Piping Circuits based on governing damage mechanism and evaluate the probability of failure and consequence of failure based on the external/internal inspection reports received from site.
- Assists with routine monitoring programs and Risk Based Inspection (RBI), corrosion rate, inspection interval calculations and remaining life calculations for Piping, Heat exchangers and Pressure Vessels for mechanical integrity assurance.
- Performing integrity analysis by Quantitative approach and Qualitative approach based on data and Engineering Judgement.
- Familiar in evaluating Consequence Assessment Plan document, fire/explosive, toxic, burn, environment and public consequences for risk assessments.
- Responsible for evaluating un-mitigated risk and assigning inspection task and inspection interval for further reduction of risk to acceptable level.
- Evaluating Safety, Health Environmental risks associated for the identified damage mechanisms to minimize the overall risk from the failure of the equipment.

PRD Interval Optimization:-

- Oversaw the full lifecycle of PRDs, ensuring compliance with industry safety standards and optimizing maintenance intervals for improved operational efficiency and safety.
- Ensured all PRD maintenance activities met regulatory requirements through detailed documentation, audits, and reporting, contributing to a culture of safety and compliance.
- Streamlined PRD inspection processes through data-driven optimization, resulting in significant cost savings by reducing unnecessary inspections and focusing on high-risk devices based on predictive analytics.

QC Engineer - Piping (Contractual, via Padmavathi Industries)

HPCL Visakha Refinery | Aug 2023 – Oct 2024

Project 2: Residue upgradation facility (RUF) Shut down

- Oversaw the quality inspection and documentation for new and modified piping systems within a major refinery shutdown/maintenance window.
- Ensured 100% compliance of all piping fabrication and installation activities with ASME B31.3 and client specifications.
- Coordinated with NDT teams to verify weld quality and integrity, streamlining the handover process.

Project 1: Residue upgradation facility (RUF) |

- Executed in-process and final QC inspections for piping components, including material receiving, fit-up, and welding activities.
- Maintained comprehensive Quality Records (QR) and Test Packs for all completed sections, which were crucial for successful commissioning.
- Identified and resolved several non-conformances related to pipe support installation and material traceability

Fabrication shop (RUF)

- Incoming Inspection: Verify materials (pipes, fittings, flanges) against specifications and codes using Mill Test Certificates (MTCs).
- Traceability: Ensure material identification and storage meet standards.

Core Competencies

Mechanical Integrity (MI)

PRD Optimization

Piping Circuitization

Process Flow Diagram (PFD) Preparation

Quality Control (QC) - Piping

ASME/API Standards (e.g., API 570, API 510)

Non-Destructive Testing (NDT) Oversight

Asset Performance Management (APM)

Skills

- AutoCAD and Microsoft Office tools
- Piping isometric, Piping and circuitization, PFD creation
- PRD optimization, pressure relief device, mechanical condition, cleanliness condition, Overhaul reports.
- Root Cause Analysis, Process Development, Engineering Documentation, Problem Solving.

Achievements

- Received ***ON THE FLY*** awards for the exemplary project deliveries and for valuable contribution to Quest Global.
- Successfully achieved high Customer Satisfaction (CSAT) ratings by delivering exceptional project results and ensuring alignment with client expectations.
- Developed a VBA-based solution to automate the find & copying, and renaming of bulk files, significantly improving efficiency and earning recognition for outstanding work.
- Collaborated with another business unit to develop a solution for bulk DWG to PDF conversions using AutoCAD LISP

Education Qualification

Bachelor of Technology (B.Tech) | Mechanical Engineering *2019-2023*
CGPA- 8.61

Intermediate | MPC -Visakhapatnam *2017-2019*
CGPA- 9.78

SSC – Visakhapatnam *2016-2017*
CGPA- 9.7

Declaration

I hereby declare that the above furnished information is true to the best of my knowledge and I am confident of my ability to work.